Phase 2: Leakage-Safe Spatial and Comparable-Sale Features for Paris Apartment Valuation

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Abstract

This addendum evaluates a second-stage data-engineering pipeline for the Paris DVF apartment valuation benchmark. A typed Gold table of 143,009 transactions adds Lambert-93 distances, H3 cells, local transaction density, and historical comparable-sale statistics. Every comparable is at least 90 days older than the transaction being valued. Model and blend selection use 2024 only; 28,330 sales from 2025 form the final chronological test. A transparent correction combining 75% of the Phase 1 LightGBM estimate with 25% of a trailing local median reduces test MAE from EUR 96,981 to EUR 96,428. The paired bootstrap interval for the reduction is EUR 333–743. Full enriched LightGBM and XGBoost models do not materially improve the Phase 1 reference. The result supports conservative use of recent local evidence and argues against complexity without new building-level information.

1 Objective and contribution

Phase 1 established a chronological six-model benchmark. Phase 2 asks whether better spatial data engineering and strictly historical comparable sales add out-of-year value. The contribution is an auditable feature contract, not a new learning algorithm. Source-file hashes, cleaning counts, feature ranges, and coverage are recorded in a machine-readable quality report.

2 Data-engineering design

Five annual Paris DVF files covering 2021–2025 are cleaned at mutation level using the fixed Phase 1 cohort definition. WGS84 coordinates are retained for storage and transformed to Lambert-93 for metric distance calculations. H3 cells at resolutions 8, 9, and 10 preserve multiple spatial scales. Address, street, commune, and parcel identifiers are retained in the Gold table, while the highest-cardinality categories are not supplied directly to the models.

For each sale at date t, eligible historical observations satisfy

$$90 \leq t - t_i \leq 730 \text{ days}, \qquad d(i,t) \leq 1000 \text{ m}.$$

Counts and median prices per square metre are computed for several nested distance/time windows. Additional features include dispersion, distance/time weighted comparable values, size/room-similar comparable values, nearest prior sale distance, median comparable age, and short-versus-long local momentum. The 90-day lag approximates information availability and prevents same-period or future transactions from leaking into a valuation.

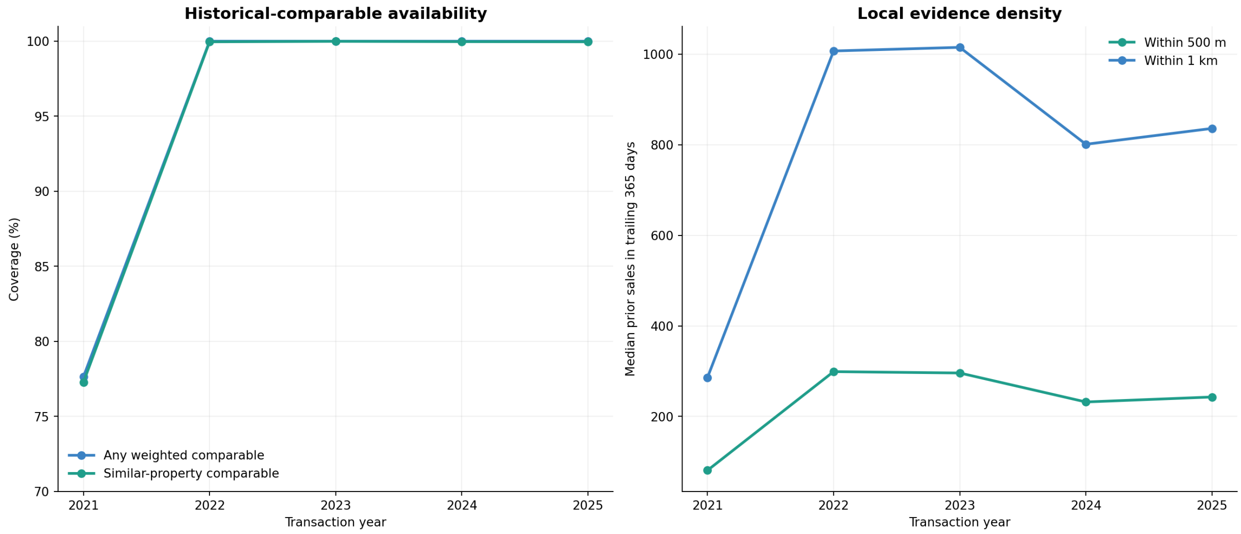

Figure 1: Coverage and density of eligible historical evidence. Lower 2021 coverage is expected because no pre-2021 source file is available.

3 Evaluation protocol

Development training uses 90,275 sales from 2021–2023. The 24,404 sales from 2024 select model configurations, target formulations, comparable definitions, and blend weights. Selected learners are refitted on 114,679 sales from 2021–2024 and tested once on 28,330 sales from 2025. The primary metric is MAE; secondary metrics are median absolute percentage error (MdAPE), R^2, and the share of estimates within 20% of the transaction value.

LightGBM and XGBoost use full spatial/comparable features with log total-price, log-price-per-square-metre, and local-residual targets. Spatial-only and comparable-only ablations test the contribution of feature groups. A six-model blend and four transparent Phase 1/comparable corrections are selected using validation MAE only.

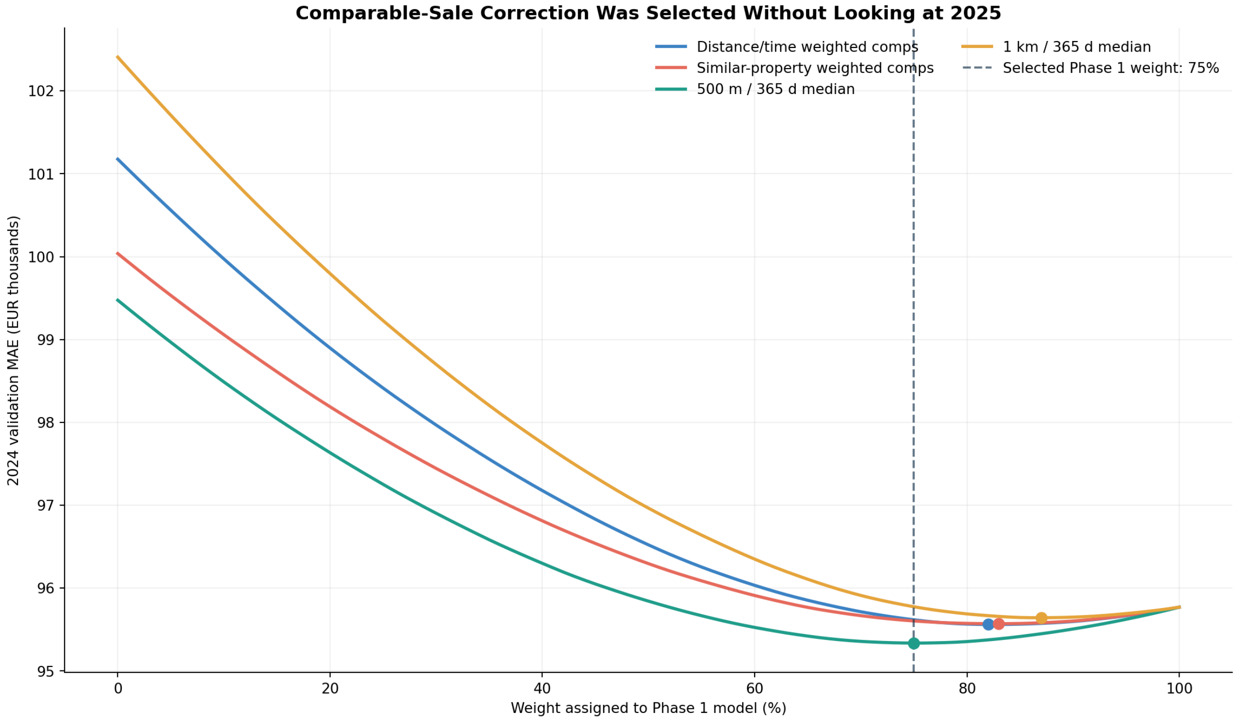

Figure 2: Validation-only selection of the comparable feature and blend weight. The selected correction assigns 75% weight to Phase 1 and 25% to the 500-metre, trailing-365-day median.

4 Results

Table 1: Selected 2025 test results. Lower MAE and MdAPE are better.

Method	MAE (EUR)	MdAPE	Within 20%
Phase 1 + comparable correction	**96,428**	**12.22%**	**71.69%**
XGBoost, total-price target	96,914	12.43%	71.18%
Phase 1 LightGBM	96,981	12.33%	71.41%
XGBoost, EUR/m^2 target	96,996	12.61%	70.73%
Six-model validation blend	97,183	12.52%	70.89%
XGBoost, residual target	97,507	12.60%	70.69%
LightGBM, spatial-only	98,232	12.72%	70.31%
LightGBM, comparable-only	98,963	12.73%	70.81%
LightGBM, full total-price	99,144	12.73%	69.89%

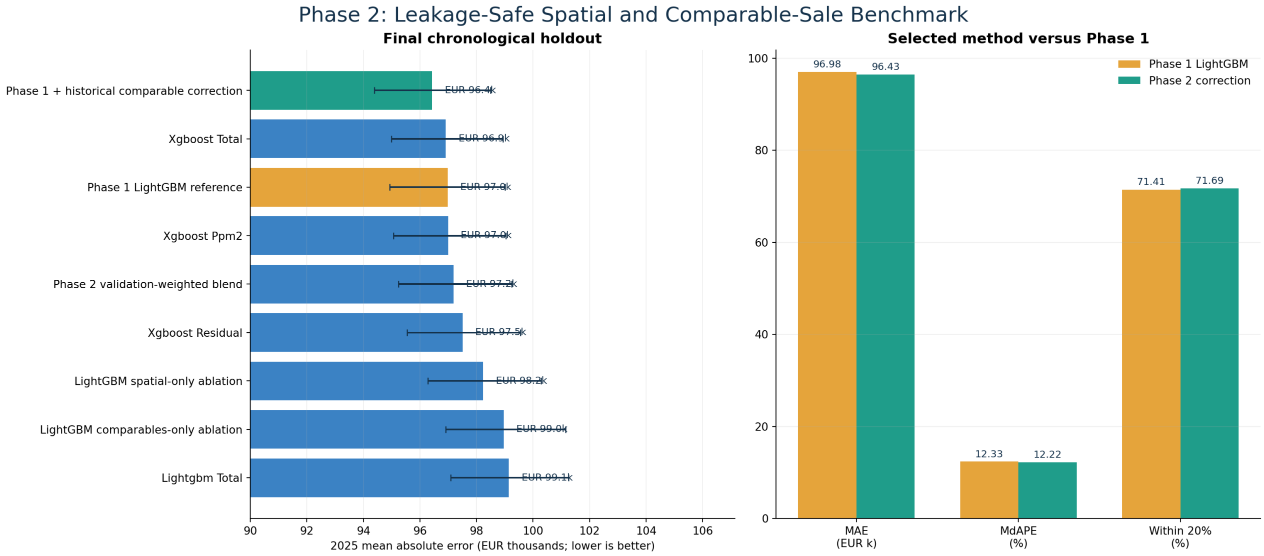

Figure 3: Chronological test accuracy. Error bars are 95% bootstrap intervals for each MAE.

The selected correction improves MAE by EUR 552 relative to Phase 1. A paired bootstrap over identical test transactions gives a 95% interval of EUR 333–743 for this reduction. MdAPE improves from 12.33% to 12.22%, and the within-20% rate increases from 71.41% to 71.69%. R^2 decreases slightly from 0.8619 to 0.8601, so the evidence is metric-dependent. The effect is statistically consistent on this cohort but modest in practical magnitude.

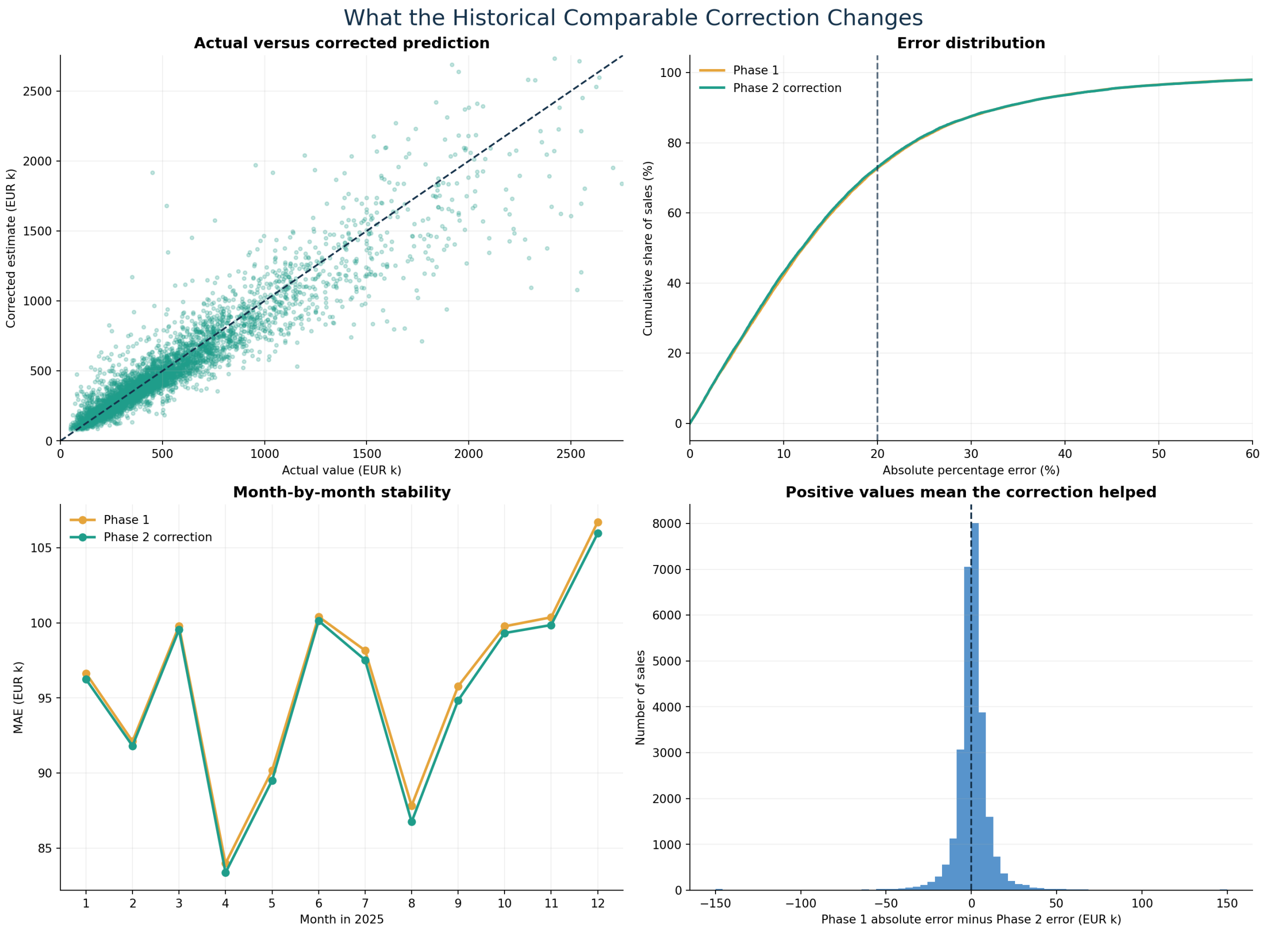

Figure 4: Prediction, error-distribution, temporal-stability, and paired-error diagnostics for the selected correction.

5 Interpretation

Adding all engineered variables to a flexible learner did not outperform the simpler Phase 1 model. The feature-gain diagnostic confirms that surface remains dominant, although XGBoost uses the local median and weighted comparable features. Importance values describe model usage, not causal effects.

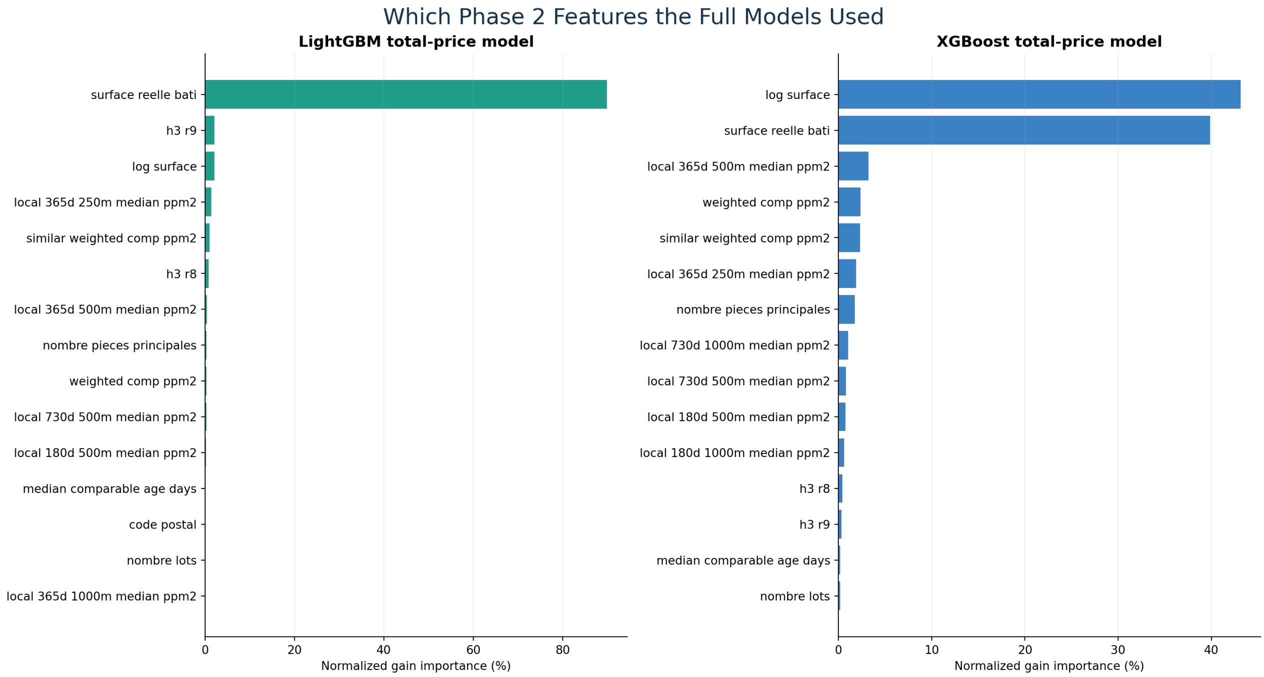

Figure 5: Gain-based feature importance for the full total-price models.

The transparent correction succeeds by shrinking the base estimate only partly toward recent local evidence. It is easier to inspect and reproduce than the six-model blend. The outcome also shows why engineering effort should be judged by chronological holdout performance rather than feature count.

6 Limitations and next step

DVF lacks condition, floor, lift, view, noise, renovation quality, detailed building state, and energy performance. Because 2025 had already been inspected during Phase 1, future claims require genuinely prospective confirmation on a later cohort. The bootstrap quantifies sampling variation within 2025, not market-regime uncertainty. The estimates are statistical indications, not certified appraisals or investment guarantees.

The recommended Phase 3 is reliable external linkage: BAN-normalized addresses, BDNB building characteristics, DPE ratings, cadastral age, and distances to transit and amenities. These additions address missing information directly and should be assessed with both chronological and spatial holdouts before exploring attention-based architectures.

7 Reproducibility

The complete build, benchmark, visualization, and single-property inference commands are documented in `reports/phase2/PHASE2_REPORT.md`. Numerical results are stored as CSV and JSON, and the Gold table is Parquet with its source hashes and feature contract recorded separately.